

HARDIK ARORA

California (Open to relocation) / +1 (517)-391-6994 / hardikarora5520@gmail.com / [LinkedIn](#) / [GitHub](#)

EDUCATION

Michigan State University

Aug 2024 – Dec 2025

Master of Science in Computer Science (GPA: 3.5 / 4.0)

MS Ramaiah University of Applied Sciences

Aug 2018 – May 2022

Bachelor of Technology in Computer Science (GPA: 3.7 / 4.0)

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, C#, JavaScript, TypeScript, Go

Frontend Development: React, Angular, Lit.js, Next.js, Micro Frontends

Backend & APIs: Node.js, Spring Boot, Flask, REST APIs, Fast API, Service-Oriented Architecture

Databases: PostgreSQL, MySQL, MongoDB, Firebase

Cloud & DevOps: AWS (EKS, ECR, CloudWatch, Sagemaker, EC2), Azure (OpenTelemetry, AKS, Key Vault), GCP (Batch, GKE), Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI

Testing & Engineering Tools: Jest, Cypress, Git, Jira, Postman, Dynatrace, Splunk

AI & ML Tools: LangChain, LangGraph, Langsmith, Elasticsearch, HuggingFace, Vector Database (Chroma, Pinecone), Scikit-learn, MCP, PyTorch

EXPERIENCE

Software Engineer Intern / Rocket Mortgage

May 2025 – Aug 2025

- Built **REST APIs** in **Java** and **Spring Boot** for the Identity Platform team, shipping **authentication and authorization** features end to end across the service layer, BFF, and Angular frontend.
- Built an **internal MCP server** integrating **LangChain** with **Auth0**, **Dynatrace**, and **Splunk APIs** to help on call engineers automate identity incident triage, reducing auth related postmortem preparation time from hours to minutes and strengthening fault tolerance across the auth stack.

Research Assistant / Basso Lab – Michigan State University

Sep 2024 – April 2026

- Built a Python-based **ETL** and **computer vision** pipeline to predict plant yields by extracting canopy cover from field images and integrating thermal time (GDD) data, achieving **~80%** prediction accuracy across multiple growing cycles.
- Migrated DSSAT crop simulation workloads from **Kubernetes** to **Google Cloud Batch**, improving scalability and cutting compute costs by **30%**.
- Automated **CI/CD pipelines** for MI Digital Farm using **GitLab CI** and **Azure**, reducing manual release effort by **60%**.
- Built a **full-stack web application** with **React** and **Flask**, integrating the **SALUS crop simulation model** and **Google Maps API** to deliver real-time geospatial insights, with **Swetrix** and **Azure OpenTelemetry** enabling end-to-end observability.

Associate Software Engineer / Conga

Jan 2022 – Aug 2024

- Architected and scaled the **Conga Component Library (CCL)** using **Lit.js**, delivering reusable UI components adopted by **40+ production applications**, reducing duplicated front-end development effort by **70%**.
- Transformed the UI platform through a **micro-frontend architecture**, which enabled independent feature deployments, improved scalability, and supported parallel development across engineering teams.
- Strengthened engineering quality and CI reliability by implementing **automated linting**, **unit testing**, and **quality gates within CI pipelines**, preventing regressions and ensuring consistent standards across all consuming applications.

Software Engineer Intern / Moody's Analytics

May 2021 – Jul 2021

- Enhanced frontend modules of the **Deal Structuring Model** using **AngularJS**, implementing reusable components, improving UI responsiveness, and streamlining complex financial workflows used by business analysts.
- Debugged and resolved **backend service and data pipeline issues** by analyzing logs, optimizing service logic, and refining database queries, resulting in **25% improvement in API response times**.

PROJECTS

AI PR Review Agent [Link](#) | Java, Springboot, Langgraph4j, React, OpenAI, Grafana, Prometheus

Feb 2026– April 2026

- Built an autonomous PR review agent using **LangGraph4j** and **GPT-4o** that triggers on **GitHub webhooks**, analyzes code diffs for quality and **OWASP** security issues, and posts structured review comments directly on the pull request.
- Designed a full stack system with a **Spring Boot** backend featuring **JWT authentication** and a **React** dashboard for browsing reviews and contributor trends, deployed on Railway with Prometheus and Grafana observability.

PR Reviewer Distillation [Link](#) | Python, Modal, Unsloth(QLoRA), AWQ, vLLM, HuggingFace

April 2026– May 2026

- Distilled **GPT-4o** into **Qwen 2.5 Code 7B Instruct** via **QLoRA** on **Modal (L40S)**, training on 73 PRs scraped from popular Python repos (fastapi, pydantic, flask, httpx, sqlmodel); merged adapter, **AWQ-quantized to 4-bit** (~4 GB), pushed to **Hugging Face** as a public model for under **\$8 total compute cost**.
- Deployed the **quantized 7B PR-review** model using **vLLM** and **Modal** with **OpenAI-compatible inference APIs** and **scale-to-zero** serving; achieved nearGPT-4o performance (**2.38 vs 2.62**) on hand-labeled code review evaluations.

CERTIFICATIONS

- Microsoft Certified Azure Fundamentals (AZ-900)
- AWS Cloud Foundations
- Full-Stack Development – IIT Kanpur